Quiz 2

05/13/2026

Question 1 (20 points)

Assume an input tensor of size 100×28×28 $(D \times W \times H)$. A convolutional layer uses 200 kernels, each with spatial size 3×3. Assume a stride of 1 and zero-padding of 1. What is the dimension of the output tensor?

A. $100 \times 15 \times 15$
B. $100 \times 27 \times 27$
C. $200 \times 28 \times 28$
D. $200 \times 15 \times 15$

Question 2 (20 points)

Assume an input tensor of size 100×28×28 $(D \times W \times H)$. A convolutional layer uses 200 kernels, each with spatial size 3×3. Assume each kernel spans the full input depth and the layer includes one bias term per kernel. How many learnable parameters does this convolutional layer have?

A. $200 \times 3 \times 3 = 1{,}800$
B. $100 \times 3 \times 3 + 200 = 1{,}100$
C. $200 \times 100 \times 3 \times 3 = 180{,}000$
D. $200 \times 100 \times 3 \times 3 + 200 = 180{,}200$

Question 3 (20 points)

Assume a video clip is represented as an input tensor of size 3×16×112×112 $(C \times T \times H \times W)$, where 3 is the number of color channels, 16 is the number of frames, and 112×112 is the spatial size. A 3D convolutional layer uses 64 kernels, each with size 3×3×3 $(T \times H \times W)$. Assume stride 1 and padding 1 in all three dimensions. What is the dimension of the output tensor?

A. $3 \times 16 \times 112 \times 112$
B. $64 \times 16 \times 112 \times 112$
C. $64 \times 14 \times 110 \times 110$
D. $3 \times 14 \times 110 \times 110$

Question 4 (40 points)

In self-supervised contrastive learning, each original image is randomly augmented twice to create two different views.

Suppose we have two original images, Image A and Image B. After data augmentation, we obtain four views:

$$A_1,\ A_2,\ B_1,\ B_2$$

The cosine similarity scores between different views are given below:

Pair	**Cosine similarity**
$sim(A_1, A_2)$	0.8
$sim(A_1, B_1)$	0.3
$sim(A_1, B_2)$	-0.2
$sim(A_2, B_1)$	-0.1
$sim(A_2, B_2)$	0.4
$sim(B_1, B_2)$	0.7

Assume the temperature parameter is:

$$\tau = 1$$

For an anchor (reference) view i, the contrastive loss is:

$$L_i = -\log \frac{\exp\left(sim(i, i^+)/\tau\right)}{\exp\left(sim(i, i^+)/\tau\right) + \sum_{j \in \mathcal{N}(i)} \exp\left(sim(i, j)/\tau\right)}$$

where:

- i is the anchor view.
- i^+ is the positive view from the same original image.
- $\mathcal{N}(i)$ is the set of negative views from other original images (i.e., $\mathcal{N}(i)$ does not include the positive i^+).
- $sim(\cdot,\cdot)$ is cosine similarity.
- $\exp(x) = e^x$, where $e \approx 2.718$.
- log denotes the natural logarithm (base *e*), i.e., log(x) = ln(x).

Answer the following questions:

1. For each anchor view, identify its positive sample and negative samples. (10 points)

2. Write the contrastive loss expression for each anchor: (10 points)

$$L_{A_1},\ L_{A_2},\ L_{B_1},\ L_{B_2}$$

3. Compute the approximate value of L_{A_1}. (10 points)

4. Write the final full-batch contrastive loss by averaging all anchor losses. (10 points)